

1. What dataset did you use? How many LIVE and SPOOF samples total?

Answer:

The project used the **SOCOFing (Sokoto Coventry Fingerprint) dataset** as the source of LIVE fingerprint images. The dataset contains approximately **6000 real fingerprint images** collected from **600 subjects**.

For the SPOOF class, **40 real presentation attack images** were manually collected by capturing printed and screen-displayed fingerprint images. To improve class diversity and address dataset imbalance, an additional **800 realistic spoof images** were generated using data augmentation techniques such as perspective transformation, motion blur, Gaussian blur, brightness variation, JPEG compression, screen-line simulation, reflections, and noise.

The final dataset used for model training consisted of:

Class	Images
LIVE	6000
SPOOF	840

For balanced training, the dataset was sampled and split into:

Split	LIVE	SPOOF
Training	672	672
Validation	84	84
Testing	84	84

2. Which CNN backbone did you choose and why?

Answer:

The project uses **MobileNetV3-Small** as the backbone convolutional neural network.

MobileNetV3-Small was selected because it is specifically designed for lightweight and efficient image classification while maintaining high accuracy. It is optimized for mobile and edge devices, making it suitable for real-time fingerprint liveness detection.

Transfer learning was applied by loading the ImageNet pre-trained MobileNetV3-Small model and replacing its final classification layer with a new fully connected layer containing two output neurons representing the LIVE and SPOOF classes.

The advantages of MobileNetV3-Small include:

- Lightweight architecture
- Fast inference

- Low computational cost
- High classification accuracy
- Suitable for real-time biometric applications

3. What threshold did you set? What is APCER at BPCER = 3%?

Answer:

Threshold calibration was performed by sweeping the decision threshold from **0.00** to **1.00** with a step size of **0.01** using the validation dataset.

The assignment specifies selecting a threshold that satisfies **BPCER $\leq$ 3%**.

The selected operating threshold was:

Threshold = 0.01

At this operating point:

- **BPCER = 0.00%**
- **APCER = 0.00%**

Since the obtained BPCER is **0.00%**, it satisfies the required condition of **BPCER $\leq$ 3%** while achieving zero spoof acceptance.

4. What is your EER?

Answer:

The Equal Error Rate (EER) represents the operating point where APCER and BPCER become equal.

The evaluation results showed:

- **EER = 0.00%**
- **EER Threshold = 0.01**

This indicates excellent separation between genuine and spoof fingerprint images on the prepared dataset.

5. What attack types would fool your model? (printed photo vs. screen replay vs. 3D mold — which is hardest?)

Answer:

The current model was trained using genuine fingerprint images together with presentation attacks including printed photographs, screen replay attacks, and realistic augmented spoof samples.

The model is expected to perform well against:

- Printed fingerprint photographs
- Screen replay attacks
- Camera recaptured fingerprint images

However, more advanced presentation attacks may still challenge the model. These include:

- High-quality silicone fingerprint molds
- Gelatin or latex fingerprints
- 3D printed fingerprint replicas
- Spoofs created using unseen materials or different sensors

Among these, **high-quality 3D molds and silicone fingerprints are expected to be the most difficult to detect**, as they more closely resemble the texture and optical properties of real fingers.

Improving robustness against such attacks would require collecting additional spoof datasets representing those attack types.

6. Why do we fix BPCER and not APCER when choosing the operating threshold?

Answer:

In biometric authentication systems, rejecting legitimate users has a direct impact on user experience. Therefore, biometric systems typically limit the maximum acceptable Bona-fide Presentation Classification Error Rate (BPCER) and then minimize the Attack Presentation Classification Error Rate (APCER) at that operating point.

By fixing **BPCER $\leq$ 3%**, the system ensures that genuine users are rarely rejected while still maximizing protection against spoof attacks.

This strategy provides a practical balance between usability and security and is commonly adopted in biometric presentation attack detection systems.

7. If you had 3 months and a proper dataset, what would you do differently?

Answer:

Given additional development time and access to larger presentation attack datasets, several improvements could be implemented.

These include:

- Collecting real presentation attack samples from multiple fingerprint sensors.
- Including silicone, gelatin, latex, and 3D printed spoof fingerprints.
- Training using larger publicly available fingerprint PAD datasets.
- Performing cross-dataset evaluation to improve generalization.
- Fine-tuning additional CNN architectures such as EfficientNet and ConvNeXt.
- Applying advanced augmentation strategies and domain adaptation techniques.
- Optimizing the model using ONNX or TensorRT for deployment on embedded and mobile devices.
- Extending the application to support real-time webcam-based fingerprint liveness detection.

These improvements would increase the robustness and practical applicability of the fingerprint presentation attack detection system in real-world biometric authentication scenarios.

